

Ayan Majumdar

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About

- 📌 Doctoral Researcher at MPI-SWS, specializing in Trustworthy AI and AI Safety. I bridge the gap between theoretical research in fairness, causality, and explainability and the practical deployment of safety-compliant Automated Decision-Making and Generative AI systems.

Research Interests

- Trustworthy AI 📌 AI Safety, Fairness, Explainability, Causality
- Deep Learning 📌 Generative AI, Large language models, Vision language models

Education

- 2021 – 2025 (expected) 📌 **Ph.D. in Computer Science**
MPI-SWS and Saarland University, Saarbrücken, Germany
Area: *Deep Learning, Generative AI, Fairness, Explainability*
Advisor: Prof. Dr. Krishna P. Gummadi, Prof. Dr. Isabel Valera
Contact: [gummadi\[at\]mpi-sws.org](mailto:gummadi[at]mpi-sws.org), [ivalera\[at\]cs.uni-saarland.de](mailto:ivalera[at]cs.uni-saarland.de)
- 2017 – 2021 📌 **M.Sc. in Computer Science**
Saarland University, Saarbrücken, Germany
Thesis: *Generating Counterfactuals for Causal Fairness*
Outline: *Deep generative models and their implicit assumptions in generating counterfactuals from observed data in the context of fairness.*
Advisor: Prof. Dr. Krishna P. Gummadi, Prof. Dr. Isabel Valera
GPA: 1.2/1.0 (German Scale)
- 2011 – 2015 📌 **B.Tech. in Electronics & Communication**
Heritage Institute of Technology, Kolkata, India
Project: *Automated traffic detection using image processing*
Outline: *Utilize blob detection techniques for detecting traffic from video sequences.*
Supervisor: Prof. Anindya Sen
GPA: 8.8/10.0

Publications

Conference Proceedings

- 1 **Majumdar, A.**, & Valera, I. (2024). CARMA: A practical framework to generate recommendations for causal algorithmic recourse at scale. In *2024 ACM Conference on Fairness, Accountability, and Transparency* (pp. 1745–1762). 📄 Paper 🔄 Code.
- 2 Nanda, V., **Majumdar, A.**, Kolling, C., Dickerson, J. P., Gummadi, K. P., Love, B. C., & Weller, A. (2023). Do invariances in deep neural networks align with human perception? In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 37, pp. 9277–9285). 📄 Paper 🔄 Code.

- 3 Rateike*, M., **Majumdar***, A., Mineeva, O., Gummadi, K. P., & Valera, I. (2022). Dont Throw it Away! The Utility of Unlabeled Data in Fair Decision Making. In *2022 ACM Conference on Fairness, Accountability, and Transparency* (pp. 1421–1433).  Paper  Code.

Archival Pre-prints

- 1 **Majumdar***, A., Kanubala*, D. D., Gupta, K., & Valera, I. (2026). A causal framework to measure and mitigate non-binary treatment discrimination. *arXiv preprint arXiv:2503.22454*. Accepted to AAAI 2026  Paper  Code.
- 2 **Majumdar**, A., Chen, F., Li, J., & Wang, X. (2025). Evaluating llms for demographic-targeted social bias detection: A comprehensive benchmark study. *arXiv preprint arXiv:2510.04641*.  Paper.

Work Experience

- Aug. 2024 – Feb. 2025  **Ph.D. Research Intern**
Huawei Research TTE-DE Lab, München, Germany
Role: Developed a benchmark using LLMs to detect demographic-targeted social bias in large datasets, documented in the preprint Evaluating LLMs for Demographic-Targeted Social Bias Detection. Conducted research surveys to support AI governance and alignment with regulations like the EU AI Act.
- Oct. 2019 – Mar. 2021  **Research Assistant**
Max Planck Institute for Software Systems, Saarbrücken, Germany
Project: *Exploring bias and fairness with deep generative models*
Role: Led project regarding exploration of bias in deep generative models for facial image data; designed core methodologies and evaluations.
Supervisor: Prof. Dr. Krishna P. Gummadi
- Apr. 2018 – Mar. 2019  **Research Assistant**
SFB1102, Saarland University, Saarbrücken, Germany
Project: *Mutual Intelligibility in Slavic Languages*
Role: Engineered automated data processing pipelines to collect and clean large-scale textual datasets for Machine Translation experiments. Developed and deployed web-based user studies to evaluate mutual intelligibility in Slavic languages.
Supervisor: Prof. Dr. Dietrich Klakow
- Jul. 2015 – Aug. 2017  **Systems Engineer**
Infosys Ltd., Bengaluru, India
Role: Optimized and validated Session Initiation Protocol (SIP) and VoIP functionalities within session border controllers. Recognized with the Infy Insta Award (2017) and Spot Award (2016) for critical project contributions

Technical Skills

Programming, Packages and Frameworks

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|------------------|--|
| Languages |  Python, R, Java, C, C++ |
| Machine Learning |  PyTorch, Lightning, NumPy, Scikit-learn, Pandas, Jupyter |
| Trustworthy ML |  CleverHans, Foolbox, robustness, AIF360, fairlearn |
| LLMs |  transformers, vLLM, FAISS, sentence-transformers, Scipy |

Technical Skills (continued)

Others  Latex, Optuna, Seaborn, Matplotlib

Teaching Assistance

- Summer 2024  **Machine Learning (Core Lecture)**, Saarland University
- Summer 2021  **Machine-Assisted Decision Making (Seminar)**, Saarland University
- Summer 2019  **Statistical Natural Language Processing (Advanced Lecture)**, Saarland University

Mentoring Experience

- May 2024 – July 2024  Co-mentored Pranav Nyati (B.Tech. student from IIT-Kharagpur) during a DAAD summer internship on algorithmic recourse using swarm algorithms.
- Sept. 2025 – Jan. 2026  Co-mentored Lena Marie Budde (M.Sc. student from Saarland University) in a joint project and Master thesis.

Talks and Posters

- 2024  **Scaling Up Causal Algorithmic Recourse with CARMA**
ACM FAccT 2024 (Rio de Janeiro), EWAF 2024 Poster Session (Mainz)
- 2022  **Don't Throw it Away! The Utility of Unlabeled Data in Fair Decision Making**
SAP Innovation Sessions, MPI-IS (Tübingen), MILA Quebec (Montreal), ACM FAccT 2022 (Seoul)

Online Certifications

- Coursera 
 - Algorithms:** Algorithmic toolbox, Data structures, Graph algorithms, String algorithms
 - Machine Learning:** Foundations, Regression, Deep Learning: Sequence Models

Academic Activities

- Program Committee  FAccT 2024-25, ICLR 2023-25, ICML 2023-24, ECML 2023, NeurIPS 2023/25, EWAF 2023/25, WebConf-Web4Good 2026, AAAI 2025-26

Language Skills

Native/Fluent		English		Bengali	
Proficient/Intermediate		Hindi		Deutsch	

Achievements

-  Infy Insta award (2017) and Spot award (2016) for significant contributions to the project, Infosys, India.